
ENGAGEMENT OBJECTIVES

- Accuracy: Rebuild the forecasting model using statistically rigorous methods that incorporate multiple demand drivers — new customer bookings, marketing spend, and credit risk variables
- Granularity: Produce forecasts at state × product × channel level, consistent with operational reporting requirements
- Reliability under data constraints: Ensure robust forecasts are available even where historical data is limited — enabling new product and new geography planning
- Scenario capability: Enable what-if modeling for launching new products in existing states, or existing products in new states
- Scalability: Python-based architecture that can absorb new states, products, and variables without structural rework
- Technology integration: Partner with client technology teams to implement and operationalize the solution in the client's environment

SCOPE OF WORK

Phase 1: Core forecast rebuild

Phase 1 rebuilds the non-new customer booking forecast from scratch. The focus is on producing an accurate, well-documented forecast file — not a platform or interface.

Objective: Predict number of non-new customer bookings at state × product × channel level through end of December 2027

Inputs: Prior new customer bookings (primary driver), marketing spend by channel, and any additional variables identified during the engagement that materially improve forecast accuracy

Methodology: Statistical regression approach incorporating multiple predictors to move beyond the current single-factor vintage model. Specific technique (Ridge regression, distributed lag, or alternative) to be confirmed following exploratory data analysis

Deliverable: A forecast file containing predicted non-new bookings by state, product, and channel through December 2027, accompanied by methodology documentation

Dependency: Forecast delivery is conditional on timely receipt of agreed input data from the client. Timeline impact of data delays will be assessed jointly

Phase 1 does not include a Streamlit interface or end-user platform — those are Phase 2 scope

Phase 2 — Platform, enhancement and implementation

Phase 2 extends the Phase 1 model across five workstreams, to be scoped and priced following Phase 1 delivery and acceptance.

Workstream 1 — Reliability under limited data: Develop techniques (e.g. look-alike state borrowing, Bayesian shrinkage, pooled estimation) to ensure reliable forecasts are available even where a state, product, or channel has short or sparse history. This is a prerequisite for new launch planning

Workstream 2 — What-if scenario modelling: Build capability to model launch scenarios: a new product entering an existing state, or an existing product entering a new state. Scenarios will generate like-for-like forecasts using the statistical structure of comparable lanes

Workstream 3 — Additional variable integration: Incorporate credit risk scores (mean, 25th, and 75th percentile distributions), funded amount, and any other variables agreed during Phase 1 that improve forecast accuracy

Workstream 4 — Python platform and end-user interface: Deliver a Python-based front-end (Streamlit or equivalent) enabling non-technical team members to run forecasts, configure scenarios, and track actuals versus forecast without Python expertise

Workstream 5 — Technology implementation: Partner with the client's technology function to deploy the solution in the client's technology environment.

The consulting team's role is bounded to two deliverables: (i) a technical requirements document specifying model inputs, outputs, dependencies, and architecture sufficient for an independent build; and (ii) a clean, fully commented Python script intended as a reference implementation and guidance document.

The technology team retains full ownership of the production script and is expected to rewrite it from the ground up using the above materials. The consulting team will define the algorithm will not be responsible for the technology team's implementation. To protect both parties, technology engagement is limited to a maximum of twelve hours of sessions covering any initial discussions, and post Phase 2 for document and script delivery and lastly for follow-up Q&A. After twelve hours the consulting team's obligations under this workstream are fulfilled. Any engagement beyond this is out of current scope and subject to mutually agreed commercial arrangement.

OUT OF SCOPE

The following are explicitly excluded from both phases unless agreed in writing by both parties:

- Forecasting for new customers — this engagement covers non-new customers only
- Data extraction, cleansing, or pipeline build from source systems — client to supply data in agreed format
- Real-time data integration or automated model retraining pipelines
- Ongoing model monitoring or maintenance post-implementation — to be addressed in a separate support agreement if required
- States, products, or channels not currently in scope unless explicitly added via written change request